

Foundations of Artificial Intelligence

M. Helmert
C. Büchner

University of Basel
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Exercise Sheet 11

Due: May 17, 2026

Please note the submission rules at the end of the exercise sheet.

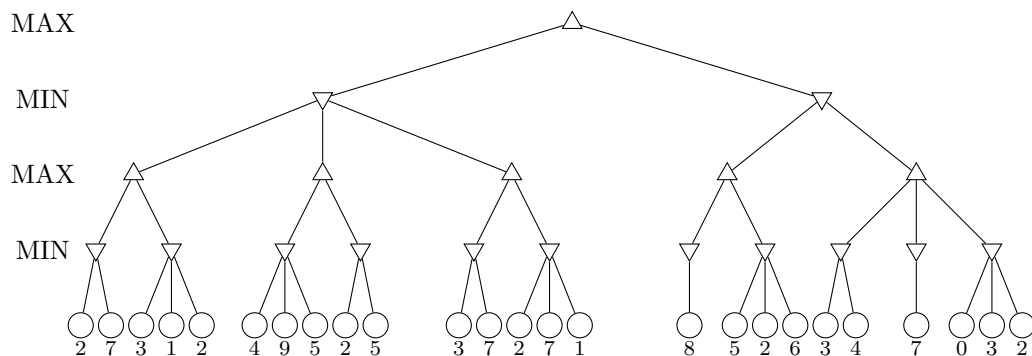
Exercise 11.1 (2+1 marks)

Consider the game *Nim* (<https://en.wikipedia.org/wiki/Nim>). The game starts with $n > 0$ stacks of coins where $c_i \geq 0$ denotes the number of coins on stack $1 \leq i \leq n$. Two players MAX and MIN alternate choosing a nonempty stack i and removing between 1 and c_i coins from it. Whichever player removes the last coin (leaving all stacks empty) wins the game.

- (a) Formalize the game *Nim* with two stacks $c_1 = 2$ and $c_2 = 1$ as a zero-sum game where MAX acts first.
- (b) Draw the game tree for the game defined in part (a). Annotate terminal nodes with their utility.

Exercise 11.2 (1.5+0.5+1.5+0.5 marks)

Consider the following game tree T :



- (a) Use the minimax algorithm to annotate T .
- (b) According to the values computed in part (a), how does the game play out if both players play optimally?
- (c) Use alpha-beta search for T . Consider successor nodes from left to right. Annotate all considered nodes with the returned value as well as the (last) alpha and beta values, and show which parts of the tree are pruned (e.g., by drawing cut lines through edges which lead to subtrees that are not considered).
- (d) How many nodes does alpha-beta search consider? Compare the utility and best move computed by alpha-beta search as well as the number of considered nodes to those of minimax in part (a). Are the values guaranteed to be the same/different/smaller/smaller-or-equal?

Exercise 11.3 (1 mark)

Consider the stochastic game $\mathcal{S} = \langle S, A, T, s_I, S_G, utility, player \rangle$ with

- $S = \{0, \dots, 44\} \times \{0, \dots, 27\} \times \{\top, \perp\} \times \{0, \dots, 27\} \times \{\top, \perp\}$
- $A = \{\text{roll}, \text{stop}\}$
- $T(\langle s, \text{roll}, t_r \rangle) = 1/6$ for $r \in \{1, \dots, 6\}$ for
 - positions $s = \langle k, v_x, a_x, v_y, a_y \rangle$ with $player(s) = \text{MAX}$ and moves $t_r = \langle k+1, v_x+r, a'_x, v_y, a_y \rangle$ where $a'_x = \perp$ iff $v_x + r > 21$.
 - positions $s = \langle k, v_x, a_x, v_y, a_y \rangle$ with $player(s) = \text{MIN}$ and moves $t_r = \langle k+1, v_x, a_x, v_y+r, a'_y \rangle$ with $a'_y = \perp$ iff $v_y + r > 21$.

$T(\langle s, \text{stop}, t \rangle) = 1$ for

- positions $s = \langle k, v_x, a_x, v_y, a_y \rangle$ with $player(s) = \text{MAX}$ and moves $t = \langle k+1, v_x, \perp, v_y, a_y \rangle$.
- positions $s = \langle k, v_x, a_x, v_y, a_y \rangle$ with $player(s) = \text{MIN}$ and moves $t = \langle k+1, v_x, a_x, v_y, \perp \rangle$.

$T(\langle s, a, t \rangle) = 0$ for all other combinations of $s, t \in S$ and $a \in A$.

- $s_I = \langle 0, 0, \top, 0, \top \rangle$
- $S_G = \{ \langle k, v_x, a_x, v_y, a_y \rangle \in S \mid a_x = a_y = \perp \}$
- $utility(\langle k, v_x, a_x, v_y, a_y \rangle) = \begin{cases} 1 & \text{if } v_y < v_x \leq 21 \text{ or } v_x \leq 21 < v_y \\ -1 & \text{if } v_x < v_y \leq 21 \text{ or } v_y \leq 21 < v_x \\ 0 & \text{otherwise} \end{cases}$
- $player(\langle k, v_x, a_x, v_y, a_y \rangle) = \begin{cases} \text{MAX} & \text{if } a_x = \top \text{ and } (k \text{ is even or } a_y = \perp) \\ \text{MIN} & \text{if } a_y = \top \text{ and } (k \text{ is odd or } a_x = \perp) \end{cases}$

Describe the rules of this dice game in natural language. What are the players trying to do?

Exercise 11.4 (2 marks)

Provide pseudocode for the expectiminimax algorithm (Chapter G5) along the lines of pseudocode for minimax provided in Chapter G2.

Hint: Recall that $\text{SUCC}(p)$ in a non-stochastic game $\mathcal{S} = \langle S, A, T, s_I, S_G, utility, player \rangle$ denotes the set of moves applicable in position $p \in S$ and the corresponding successor positions according to T , i.e., $\text{SUCC}(p) = \{ \langle move, p' \rangle \mid \langle p, move, p' \rangle \in T \}$. If $\mathcal{S}$ is a stochastic game instead, T is a *transition function* that is well-defined for all $\langle p, move \rangle$. For a given pair of positions $p, p' \in S$ and a *move* $\in A$, $T(p, move, p')$ denotes the probability of obtaining p' from applying *move* in p : if *move* is not applicable in p , then $T(p, move, p') = 0$ for all p' , otherwise T with fixed p and *move* is a probability distribution over S .

For this exercise, assume there are functions $\text{APPLICABLE}: S \rightarrow 2^A$ and $\text{PROB}: S \times A \rightarrow S \times \mathbb{R}$ mapping states to sets of applicable actions and state-action pairs to the probability distribution of possible outcomes, respectively. They are formally defined as follows:

$$\begin{aligned} \text{APPLICABLE}(p) &= \{ move \in A \mid \exists p' \in S : T(p, move, p') > 0 \} \\ \text{PROB}(p, move) &= \{ \langle p', T(p, move, p') \rangle \mid p' \in S, T(p, move, p') > 0 \} \end{aligned}$$

Submission rules:

- Submit one solution in groups of up to two students, stating the names of all group members in the submission. All group members are equally responsible for all submitted exercises.

- Create a single A4 PDF file for all non-programming exercises. Solutions in L^AT_EX are preferred, but clearly legible scans in A4 with reasonable contrast and file size are accepted as well.
- In case there are programming exercises, upload a ZIP file (ending .zip, .tar.gz or .tgz; *not* .rar or anything else) including your PDF and those code textfiles required by the exercise. Put your names in a comment on top of each code file. Make sure your code compiles and test it. Code that does not compile or which we cannot successfully execute will not be graded.
- Include a statement on scientific integrity at the end of your submission (see below).

Statement on scientific integrity:

- All submitted solutions must be the product of your own work.
- If your written solutions were produced using any help (e.g., from peers or a large language model), list in what way these people or tools contributed to the solutions.
- We do not mark submissions that lack a statement on scientific integrity.
- Examples:
 - “We solved all exercises on our own.”
 - “We used ChatGPT to explain exercise 7 which we did not understand.”
 - “We had no time to solve this exercise sheet and copied all solutions from our friends.”
(If this is the case, thanks for your honesty but this is plagiarism, results in a warning and no marks. It is preferable to submit no solution at all.)